

Question description

A company has n employees with certain salaries. Your task is to keep track of the salaries and process queries.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq p_i \leq 10^9$
- $1 \leq k \leq n$
- $1 \leq x \leq 10^9$
- $1 \leq a \leq b \leq 10^9$

Input

The first input line contains two integers n and q : the number of employees and queries. The employees are numbered $1, 2, \dots, n$.

The next line has n integers $p_1, p_2, \dots, p_n$: each employee's salary.

After this, there are q lines describing the queries. Each line has one of the following forms:

- $! k x$: change the salary of employee k to x
- $? a b$: count the number of employees whose salary is between $a \dots b$

Output

Print the answer to each $?$ query.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5 3
3 7 2 2 5
? 2 3
! 3 6
? 2 3

EXPECTED OUTPUT

3
2

Test Case 2

INPUT (STDIN)

5 3
31 17 12 12 15
? 12 31
! 13 6
? 2 13

EXPECTED OUTPUT

5
3

▼ Mandatory Test Cases

Test Case 1

KEYWORD

int compare(const void
*a,const void *b)

Test Case 2

KEYWORD

void update(int i,int
n,int x)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

1

Test Case 2

TOKEN COUNT

575

Test Case 3

NLOC

65

```
Tree18.c +
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAXN 400005
5
6 long long arr[MAXN];
7 long long vals[MAXN];
8 int bit[MAXN];
9 int n, q, sz;
10
11 char qtype[MAXN];
12 long long qa1[MAXN];
13 long long qa2[MAXN];
14
15 int cmp(const void *a, const void *b) {
16     long long x = *(const long long *)a;
17     long long y = *(const long long *)b;
18     return x < y ? -1 : (x > y ? 1 : 0);
19 }
20
21 int index_of(long long x) {
22     int lo = 1, hi = sz, ans = sz + 1;
23     while (lo <= hi) {
24         int mid = (lo + hi) / 2;
25         if (vals[mid] >= x) {
```

```

26         ans = mid;
27         hi = mid - 1;
28     } else {
29         lo = mid + 1;
30     }
31 }
32 return ans;
33}
34
35int upper_of(long long x) {
36     int lo = 1, hi = sz, ans = 0;
37     while (lo <= hi) {
38         int mid = (lo + hi) / 2;
39         if (vals[mid] <= x) {
40             ans = mid;
41             lo = mid + 1;
42         } else {
43             hi = mid - 1;

```

```
44     }
45 }
46 return ans;
47}
48
49void update(int i, int x) {
50     while (i <= sz) {
51         bit[i] += x;
52         i += i & (-i);
53     }
54}
55
56int query(int i) {
57     int res = 0;
```

main | CH.SC.U4CSE25225 | Tree18.c

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```
59     res += bit[i];
60     i -= i & (-i);
61 }
```

```
62     return res;
63}
64
65int main() {
66     int i;
67     char op;
68     if (scanf("%d %d", &n, &q) != 2) return 0;
69     for (i = 1; i <= n; i++) {
70         scanf("%lld", &arr[i]);
71         vals[sz++] = arr[i];
72     }
73     for (i = 0; i < q; i++) {
74         scanf(" %c", &op);
75         qtype[i] = op;
76         if (op == '!') {
77             int k;
78             long long x;
79             scanf("%d %lld", &k, &x);
```

```
80         qa1[i] = k;
81         qa2[i] = x;
82         vals[sz++] = x;
83     } else {
84         long long a, b;
85         scanf("%lld %lld", &a, &b);
86         qa1[i] = a;
87         qa2[i] = b;
88     }
89 }
90
91 qsort(vals, (size_t)sz, sizeof(long long), cmp);
92 int uniq = 0;
93 int j = 0;
94 while (j < sz) {
95     int k = j;
96     while (k < sz && vals[k] == vals[j]) k++;
97     vals[uniq++] = vals[j];
```

```

98     j = k;
99 }
100 sz = uniq;
101
102 for (i = 1; i <= n; i++) update(index_of(arr[i]), 1);
103 for (i = 0; i < q; i++) {
104     if (qtype[i] == '!') {
105         int k = (int)qa1[i];
106         update(index_of(arr[k]), -1);
107         arr[k] = qa2[i];
108         update(index_of(arr[k]), 1);
109     } else {
110         printf("%d\n", query(upper_of(qa2[i])) - query(index_of(qa1[i]) - 1));
111     }
112 }
113 return 0;
114 }

```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

5 3

3 7 2 2 5

? 2 3

! 3 6

? 2 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Tree18.exe

3

2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

5 3

31 17 12 12 15

? 12 31

! 13 6

? 2 13

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Tree18.exe

5

2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>